CropShield AI Dataset Details

Provenance, schema, preprocessing and responsible interpretation

Dataset identity and provenance

The operational sample is **data/sample/sample_crop_data.csv**; the application also uses **data/processed/processed_data.csv**. Source code marks the 1,500-row sample as synthetic test data. It is generated inside this project and has no external dataset link or verified field provenance. Do not present it as an official government or surveyed dataset.

Attribute	Value
Dataset name	CropShield AI synthetic crop case dataset
External source / URL	None; locally generated synthetic test data.
Format	CSV, comma-separated UTF-8 text
Rows	1,500 in the sample and processed dataset (verify locally if files are regenerated).
Purpose	Hybrid case retrieval, crop advisory context, model training and evaluation.
Crop labels	12 crop categories
Region labels	15 states in saved model metadata
Season labels	Kharif, Rabi, Summer, Whole Year
Raw-to-processed	Processed data adds NPK_Total, NPK nutrient ratios, Rain_Temp_Ratio and Fertilizer_Per_Area.
Missing values	The current processed file's saved admin inspection reports zero missing values; re-check after replacing inputs.
License / field consent	No external license or field-consent claim applies; generated test data.

Core fields

Field	Meaning
Crop	Crop variety label used in advisory and model features.
Crop_Year	Year field associated with the generated case.
Season	Agricultural season category.
State	Indian state/region category.
Area	Cultivated area feature.
Production	Production outcome feature.
Yield	Yield outcome used as regression target.
Fertilizer, Pesticide	Agricultural input quantities/features.
N, P, K	Soil macronutrient values.
pH, Temperature, Rainfall, Humidity	Soil and climate features.

Preprocessing and use

The preprocessing pipeline cleans and normalizes the CSV, handles feature conversions and records data quality. The processed file contains engineered ratios and per-area fertilizer values. Model training uses Crop, Season and State as

categorical variables; N, P, K, pH, temperature, rainfall, humidity, fertilizer and pesticide as numeric variables. Dataset path settings and sample values are in app/config/settings.py and data/; the files were intentionally not moved.

Limitations

Synthetic distributions do not encode real regional agronomic variation. The model's test metrics measure fit to a held-out split from generated data, not expected error in farmers' fields. Replace with appropriately licensed, quality-checked local data and retrain before operational decision-making.